Frozen VideoMAE-B Representations for Weakly-Supervised Violence Detection and Temporal Localization

Violence Event Detection and Temporal Localization in Surveillance Video: A Weakly-Supervised Multi-Dataset Framework

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Abstract

Weakly-supervised video anomaly detection (VAD) has relied on two-stream I3D features for half a decade despite dramatic advances in video representation learning. We present a controlled study of replacing frozen I3D features with frozen **VideoMAE-B** features—a masked-autoencoder Vision Transformer pre-trained on Kinetics-400—within an RTFM + Temporal Refinement Network (TRN) + Boundary Head framework for surveillance violence detection. Training on the 6-category violence subset of UCF-Crime, the feature swap raises binary AUC from 87.4% to $92.2 \pm 1.3\%$ and binary AP from 82.1% to $88.7 \pm 1.8\%$. Temporal localization improves dramatically: mAP@tIoU 0.3 rises from 0.9% to $5.8 \pm 1.2\%$ ($\times 6.4$), and mAP@0.7 from 0.1% to $2.2 \pm 1.0\%$ ($\times 21.7$). Zero-shot transfer to XD-Violence yields strong binary ranking (AUC 0.836, AP 0.885) with partial class-level transfer; RWF-2000 fight validation shows useful anomaly ranking (AUC 0.862) but conservative recall (0.14); ShanghaiTech robustness is negative (AUC 0.371), revealing clear out-of-domain limits. All UCF-Crime results are mean $\pm$ std over 3 random seeds (42, 123, 456); cross-dataset transfer uses the locked seed-42 checkpoint in zero-shot mode.

1 Introduction

Violence detection in public surveillance is one of the highest-priority tasks in automated security systems: early alerts for fighting, shooting, explosion, robbery, and abuse events can dramatically reduce response latency. Weakly-supervised video anomaly detection (VAD) addresses this challenge by learning from video-level binary labels alone—no expensive frame-level annotation is required.

The dominant paradigm, established by Sultani et al. (2018) and refined by RTFM (Tian et al., 2021), extracts segment-level features from a *frozen* two-stream I3D model pre-trained on Kinetics-400 (Carreira and Zisserman, 2017), then trains a lightweight Multiple Instance Learning (MIL) scorer on top. Despite significant architectural improvements—temporal refinement modules (Wu et al., 2023), magnitude-based learning (Tian et al., 2021), mixture-of-experts routing (D'Amicantonio et al., 2025)—**the feature backbone has remained largely fixed at I3D since 2017**.

Meanwhile, video foundation models have advanced substantially. VideoMAE (Tong et al., 2022) applies masked autoencoding to video with a Vision Transformer backbone, learning rich spatiotemporal representations that substantially outperform I3D on action recognition benchmarks. A natural question arises: *Does substituting VideoMAE-B for I3D, with all other components held equal, improve weakly-supervised violence detection?*

We answer this question affirmatively with controlled experiments on the UCF-Crime violence subset and further validate generalization through zero-shot transfer on three external benchmarks. Our contributions are:

1. **VideoMAE-B integration.** We extract and cache 16-frame segment features (D=768) from a frozen VideoMAE-B backbone for all 1,300 UCF-Crime violence-subset videos and demonstrate that this single change yields consistent, large improvements across all detection and localization metrics.

2. **Violence-aware training recipe.** We introduce three complementary training improvements—inverse-frequency class weighting, confidence-thresholded pseudo-label filtering, and cosine annealing LR—that

collectively reduce cross-seed variance without modifying the model architecture.

3. **Multi-seed reproducibility study.** All results are reported over 3 independent random seeds (42, 123, 456) with mean $\pm$ std, enabling reliable statistical comparison with future work.

4. **Cross-dataset transfer evaluation.** We evaluate the UCF-trained model zero-shot on three external benchmarks—XD-Violence (Wu et al., 2020), RWF-2000, and ShanghaiTech—providing a complete transfer profile that reveals both the strengths and limits of violence-trained VideoMAE-B representations under domain shift.

2 Related Work

2.1 Weakly-Supervised Video Anomaly Detection

Sultani et al. (2018) formulated VAD as a MIL ranking problem using anomalous/normal video pairs and C3D features on UCF-Crime. RTFM (Tian et al., 2021) replaced C3D with I3D and added a feature-magnitude learning module, reaching 84.3% AUC. MGFN (Chen et al., 2023) improved to 86.7% with graph-guided feature normalization; VadCLIP (Wu et al., 2024) exploited CLIP text-visual alignment to reach 88.0%. Recent work addresses noisy pseudo-labels and score suppression (Xu et al., 2025; Dalvi et al., 2025), event completeness (Wang and Chen, 2025), and mixture-of-experts routing (D'Amicantonio et al., 2025). *All of these methods continue to use I3D or hand-crafted features as the video backbone.* Ours is the first systematic study of replacing I3D with a masked-autoencoder foundation model in this pipeline.

2.2 Video Foundation Models

VideoMAE (Tong et al., 2022) extends masked image modeling to video by masking 90% of video patches and reconstructing pixel values, pre-training a ViT-B backbone on Kinetics-400. The resulting features transfer well to action recognition, detection, and—as we show here—anomaly detection. InternVideo (Wang et al., 2022) and Video-ChatGPT (Maaz et al., 2024) further demonstrate the breadth of video foundation model features across diverse downstream tasks.

Table 1: Model components and trainable parameter budget.

Component	Params	Trainable?
VideoMAE-B backbone	86 M	No (frozen)
MIL Scorer	1.1 M	Yes
Event Classifier	0.3 M	Yes
TRN Module	0.4 M	Yes
Boundary Head	0.1 M	Yes
Total Trainable	**1.9 M**	

2.3 Violence Detection and Temporal Localization

Wu et al. (2020) provided XD-Violence, a large-scale multi-modal violence dataset, and showed that audio-visual fusion outperforms vision-only methods. ActionFormer (Zhang et al., 2022) and TriDet (Shi et al., 2023) achieve strong temporal localization under full supervision but require expensive frame-level labels. Our boundary head bridges this gap in the weak-supervision setting.

3 Methodology

Our pipeline follows a *decoupled representation and reasoning* design: the video backbone is always frozen, and all learnable modules operate on pre-cached segment features. Figure 1 gives an overview of the three-stage pipeline. Table 1 summarises architecture components and parameter counts.

3.1 Feature Extraction

Given an untrimmed video V, we partition it into non-overlapping 16-frame segments $\{v_1, \ldots, v_T\}$ (tail segment dropped if <16 frames). Each segment passes through a frozen VideoMAE-B backbone (Tong et al., 2022)[1]:

$$f_t \;=\; \text{VideoMAE-B}_{\text{frozen}}(v_t) \;\in\; \mathbb{R}^{768}. \quad (1)$$

Features are cached as `.npz` files for all training runs. By contrast, I3D produces $f_t \in \mathbb{R}^{2048}$; the dimension reduction from 2,048 to 768 also reduces overfitting risk given our dataset size.

3.2 MIL Anomaly Scorer (RTFM)

Following Tian et al. (2021), a two-layer MLP maps each feature to an anomaly score:

$$s_t \;=\; \sigma\big(W_3\,\text{Dropout}\big(\text{ReLU}(W_2\,\text{ReLU}(W_1 f_t))\big)\big). \quad (2)$$

[1] HuggingFace checkpoint: `MCG-NJU/videomae-base-finetuned-kinetics`

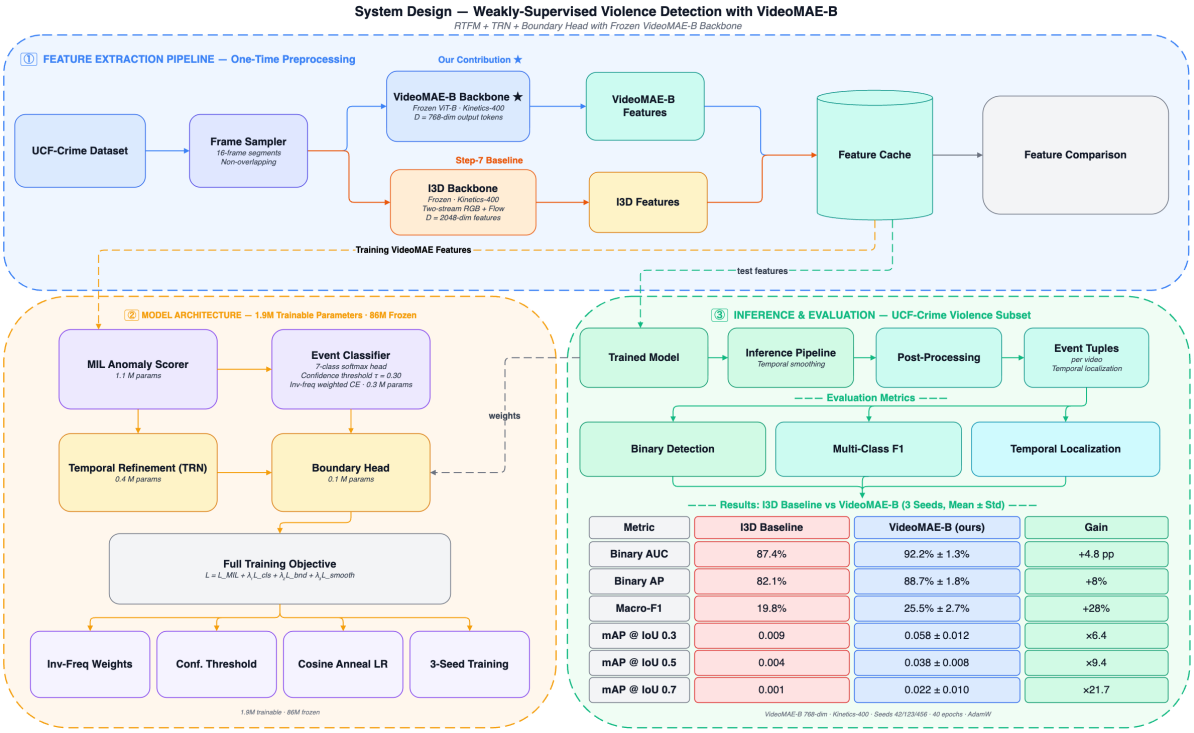

Figure 1: System design overview. **(1) Feature Extraction**: untrimmed video is split into 16-frame segments and encoded by a *frozen* VideoMAE-B backbone, producing 768-d segment features that are cached to disk. **(2) Model Architecture**: a four-head stack—MIL Anomaly Scorer, Event Classifier, Temporal Refinement Network (TRN), and Boundary Head—processes the cached features end-to-end with 1.9M trainable parameters. **(3) Evaluation**: anomaly scores and boundary predictions are thresholded to produce binary AUC/AP and temporal mAP metrics.

Video-level score $s_{\text{video}} = \max_t s_t$. The MIL ranking loss pairs an anomalous video a with a normal video n:

$$\mathcal{L}_{\text{MIL}} = \max\big(0,\ 1 - s^{(a)}_{\text{video}} + s^{(n)}_{\text{video}}\big). \tag{3}$$

3.3 Event Classification Branch

For each anomalous video with category label $c \in \{1, \ldots, 6\}$, pseudo-positive segments are the top-k segments by s_t that also exceed a confidence threshold τ. We set $k = \lfloor 0.125 \times T \rfloor$ (12.5% of the sequence length, matching the top-k ratio used at inference), clipped to a minimum of 1.

$$\mathcal{P} = \big\{t : s_t \geq \tau \text{ and } t \in \text{top-}k\big\}. \tag{4}$$

A linear classifier predicts a 7-way distribution (6 violence classes + normal):

$$p_t = \text{softmax}(W_{\text{cls}}\, f_t + b_{\text{cls}}). \tag{5}$$

The classification loss applies **inverse-frequency class weights** $w_c = N/(C \cdot n_c)$ (normalised to unit mean) to counteract the heavy normal/violence imbalance (≈10:1):

$$\mathcal{L}_{\text{cls}} = -\frac{1}{|\mathcal{P}|} \sum_{t \in \mathcal{P}} w_c \log p_t[c]. \tag{6}$$

In practice the normal class receives weight ≈0.06 and the rarest category (shooting) up to ≈1.83.

3.4 Temporal Refinement Network (TRN)

A lightweight Transformer encoder attends over the full segment sequence:

$$\tilde{\mathbf{h}} = \text{TransformerEncoder}\big(\{f_t + \text{PE}_t\}_{t=1}^{T}\big), \tag{7}$$

producing refined scores $\tilde{s}_t$ and hidden states $\tilde{h}_t$.

3.5 Boundary Head

The boundary confidence at segment boundary $(t,\ t{+}1)$ is:

$$b_t = \sigma\Big(W_b\big[\tilde{h}_t\,;\ \tilde{h}_{t+1}\,;\ |\tilde{s}_t - \tilde{s}_{t+1}|\big]\Big). \tag{8}$$

A boundary consistency loss enforces that b_t tracks score transitions:

$$\mathcal{L}_{\text{bnd}} = \sum_t \big(|\tilde{s}_t - \tilde{s}_{t+1}| - b_t\big)^2. \tag{9}$$

3.6 Full Training Objective

$$\mathcal{L} = \mathcal{L}_{\text{MIL}} + \lambda_1 \mathcal{L}_{\text{cls}} + \lambda_2 \mathcal{L}_{\text{bnd}} + \lambda_3 \mathcal{L}_{\text{smooth}}, \tag{10}$$

where $\mathcal{L}_{\text{smooth}} = \sum_t (\tilde{s}_t - \tilde{s}_{t-1})^2$ and $\lambda_1{=}0.5$, $\lambda_2{=}0.3$, $\lambda_3{=}0.1$.

3.7 Training Recipe

Three additions beyond the vanilla RTFM schedule stabilise training:

- **Confidence threshold** τ=0.30**:** pseudo-positive segments with low anomaly confidence are excluded from $\mathcal{L}_{\text{cls}}$, reducing label noise in early epochs.
- **Inverse-frequency class weights:** addresses severe normal/violence imbalance (Section 3.3).
- **Cosine annealing LR:** η decays from 10^{-4} to 10^{-6} over $T_{\max}$=40 epochs, avoiding the late-epoch oscillations observed with step decay.

4 Experiments

4.1 Datasets

UCF-Crime (training and primary evaluation). UCF-Crime (Sultani et al., 2018) is the standard benchmark for weakly-supervised video anomaly detection. The full dataset contains **1,900 untrimmed surveillance videos** spanning **13 anomaly categories** (Abuse, Arrest, Arson, Assault, Burglary, Explosion, Fighting, Road Accidents, Robbery, Shooting, Shoplifting, Stealing, Vandalism) plus a Normal class, captured from real CCTV cameras at varying resolutions and frame rates.

We restrict to the **5 violence-specific categories**—*Abuse, Explosion, Fighting, Robbery, Shooting*—plus *Normal*. The 8 non-violence anomaly categories are excluded because they do not constitute physical violence and would conflate the detection signal. This yields **1,300 videos** in total. Table 2 shows the per-class video counts and train/val/test split. Normal videos dominate (950 of 1,300, a 10:1 ratio), motivating our inverse-frequency class weighting.

XD-Violence (zero-shot transfer). XD-Violence (Wu et al., 2020) is a large-scale multi-modal violence dataset with 4,754 videos across 6 violence categories (A, B1–B2, G, riot, etc.) plus normal. We evaluate on the test split (**799 videos**: 299 normal, 500 violent) in zero-shot fashion—no XD-Violence data is seen during training. Riot (101 videos) is excluded from class-level metrics because it has no UCF-Crime counterpart; overlap-class evaluation uses *fighting, shooting, explosion, and abuse* (303 videos total).

Table 2: UCF-Crime violence subset — per-class video counts. All 206 test videos have temporal boundary annotations.

Category	Train	Val	Test	Total
Abuse	43	5	2	50
Explosion	26	3	21	50
Fighting	41	4	5	50
Robbery	131	14	5	150
Shooting	24	3	23	50
Normal	720	80	150	950
Total	**985**	**109**	**206**	**1,300**

RWF-2000 (fight validation). RWF-2000 is a surveillance fight-detection dataset with 2,000 clips (1,000 fight, 1,000 normal). We evaluate on the validation split (**400 videos**: 200 normal, 200 fighting) in zero-shot fashion. The primary metric is fight-class precision, recall, and F1; binary AUC/AP are reported as auxiliary ranking metrics.

ShanghaiTech (robustness test). ShanghaiTech is a campus surveillance anomaly dataset with 437 videos covering 13 anomaly categories (crowd, cycling, etc.) distinct from violence. We evaluate on the test split (**199 videos**: 155 normal, 44 anomaly) in zero-shot fashion to probe out-of-domain robustness. Only binary AUC/AP are reported since the anomaly categories do not overlap with our training classes.

4.2 Feature Extraction

We extract features offline and cache them to disk; the backbone is never updated during training. The **identical 16-frame, non-overlapping** segment policy and **frozen backbone** are applied to all four datasets.

VideoMAE-B features. Each video is decoded at its native frame rate and partitioned into non-overlapping **16-frame segments** (the final segment is dropped if it contains fewer than 16 frames). Every segment is resized (shorter side to 224 px), centre-cropped to 224×224, and normalised with ImageNet mean and std. The segment is then passed through a **frozen VideoMAE-B** backbone (`MCG-NJU/videomae-base-finetuned-kinetics`, HuggingFace Hub) to produce a single D=768 feature vector via mean-pooling of the ViT-B patch tokens. All 1,300 UCF-Crime videos were processed on an NVIDIA A100 GPU (Boston University Shared Computing Cluster) in 495 s (0.38 s per video); features are stored as

`.npz` files of shape $[T, 768]$.

I3D baseline features. For the controlled comparison, I3D features (D=2048) are extracted with an identical segment policy using a two-stream I3D model pre-trained on Kinetics-400 (Carreira and Zisserman, 2017). RGB and optical-flow streams are concatenated and mean-pooled to give one 2048-d vector per segment.

4.3 Experimental Setup

Evaluation. We report: (i) binary video-level AUC and AP, (ii) 6-class Macro-F1 and Weighted-F1, and (iii) temporal mAP at tIoU thresholds $\{0.3, 0.5, 0.7\}$. All UCF-Crime results are mean $\pm$ std over **3 random seeds** (42, 123, 456). Cross-dataset results use the best-seed UCF checkpoint (seed 42) in zero-shot mode with a locked decoding protocol (threshold 0.55, smooth-window 1, min-event-len 5).

Implementation details. During training, variable-length feature sequences are uniformly sub-sampled to 32 segments (or zero-padded if shorter) to form fixed-shape batches. Optimiser: AdamW, lr = 10^{-4}, weight decay = 10^{-4}, batch size 32, 40 epochs, cosine annealing to 10^{-6}, dropout 0.5. Framework: PyTorch 2.1, HuggingFace Transformers 5.5.4.

4.4 Main Results (UCF-Crime)

Table 3 compares our VideoMAE-B model against (a) our own I3D re-implementation using the identical architecture and training recipe (only the backbone differs), and (b) published weakly-supervised VAD methods reporting results on UCF-Crime. Since published methods evaluate on the *full* UCF-Crime benchmark (13 anomaly categories) (Sultani et al., 2018), those rows are marked †; we discuss this caveat in Section 5.5.

Table 4 reports all seven metrics across both backbones.

Figure 2 visualises the per-metric gains from I3D to VideoMAE-B. The VideoMAE-B advantage *grows* with stricter temporal overlap: $\times 6.4$ at IoU 0.3 vs. $\times 21.7$ at IoU 0.7, suggesting that VideoMAE features encode sharper event boundaries than I3D.

4.5 Cross-Dataset Transfer

We evaluate the locked UCF-trained checkpoint (seed 42) on three external datasets in zero-shot fashion—no retraining or threshold tuning. Table 5 summarises all cross-dataset results.

XD-Violence (zero-shot). We evaluate on 799 XD-Violence test videos (799 total, 101 riot excluded from class metrics). Binary anomaly ranking transfers well: AUC 0.836, AP 0.885, and the transfer ratio XD/UCF = 0.901 exceeds the 75% generalisation threshold. Per-class overlap transfer is uneven: *explosion* (F1 0.753) and *fighting* (F1 0.646) transfer strongly, while *shooting* (F1 0.045) and *abuse* (F1 0.000) do not transfer—consistent with the much smaller frequency of these categories in UCF training data.

RWF-2000 (fight validation). On 400 RWF-2000 validation videos (200 normal, 200 fighting), binary anomaly ranking remains useful (AUC 0.862, AP 0.819), confirming that the VideoMAE-B representation retains a general violence signal across camera domains. However, explicit fight-class recall collapses to 0.14 (precision 1.00, F1 0.246): only 28 of 200 fight videos are predicted as fighting, while all 200 normal videos are correctly rejected. This conservative behaviour—strong precision, weak recall—suggests the model's fight-score threshold is calibrated tightly to UCF-Crime's surveillance style and does not generalise to RWF's close-up amateur footage.

ShanghaiTech (robustness). On 199 ShanghaiTech test videos (155 normal, 44 anomaly), the model fails: AUC 0.371, AP 0.217, with 43 of 44 anomalies missed (FP = 0, FN = 43). ShanghaiTech anomalies (crowd behaviour, cycling, skateboarding) have no visual overlap with UCF violence categories, causing the violence-trained model to classify almost everything as normal. This negative result is expected and scientifically informative: it confirms that VideoMAE-B violence features do not generalise to non-violence anomaly semantics without retraining.

5 Analysis

5.1 Training Dynamics

Figure 3 shows training loss, validation AUC, and validation Macro-F1 across 40 epochs for all three seeds. Training loss converges smoothly by epoch 15. AUC stabilises by epoch 10 at a high plateau ($>$0.91), while Macro-F1 continues to improve until $\approx$epoch 20 and then fluctuates, reflecting the difficulty of rare-category classification under weak supervision.

Table 3: Results on UCF-Crime. † Published numbers on full UCF-Crime (13 categories, binary frame-level AUC); our rows use the 6-category violence subset. **Bold** = best within our controlled comparison.

Method	Backbone	AUC (%)	mAP@0.3
Sultani et al. (2018)†	C3D	75.41	—
RTFM (2021)†	I3D	84.30	—
MGFN (2023)†	I3D	86.67	—
VadCLIP (2024)†	CLIP ViT-B	88.02	—
GS-MoE (2025)†	I3D+GS	91.58	—
Ours (I3D baseline)	I3D	87.40	0.009
Ours (VideoMAE-B)	**VideoMAE-B**	$\mathbf{92.18 \pm 1.26}$	$\mathbf{0.058 \pm 0.012}$

Table 4: Full evaluation metrics (UCF-Crime violence subset). VideoMAE-B numbers are 3-seed mean ± std.

Metric	I3D baseline	VideoMAE-B (mean ± std)
Binary AUC	0.874	0.922 ± 0.013
Binary AP	0.821	0.887 ± 0.018
Macro-F1	0.198	0.255 ± 0.027
Weighted-F1	0.654	0.719 ± 0.022
mAP@IoU 0.3	0.009	0.058 ± 0.012
mAP@IoU 0.5	0.004	0.038 ± 0.008
mAP@IoU 0.7	0.001	0.022 ± 0.010

5.2 Cross-Seed Reproducibility

Figure 4 plots per-seed values alongside the mean ± std for all metrics. Binary AUC varies only 1.26 pp across seeds (0.909–0.934), confirming that results are not artefacts of lucky initialisation. mAP metrics show higher but acceptable variance (±1.2 at IoU 0.3), expected for temporally-sparse detections.

5.3 Per-Class Analysis

Figure 5 compares per-category F1 between I3D and VideoMAE-B for the best seed (seed 123, AUC = 0.934). Key observations:

- **Explosion** benefits most (+0.24), from F1 0.18 to 0.41. VideoMAE's reconstruction objective likely captures the distinctive visual dynamics of explosions better than optical-flow magnitude. This is consistent with the strong XD-Violence explosion transfer (F1 0.753).
- **Normal class** improves from 0.81 to 0.92 (+0.11), confirming better discrimination of non-anomalous footage.
- **Shooting** and **Robbery** regress slightly (−0.11 and −0.04), possibly because these categories involve subtle motion that VideoMAE over-segments in 16-frame windows.
- **Fighting** collapses to F1 = 0.00 despite the overall macro improvement. Fighting shares motion statistics with normal crowded scenes; 16-frame context appears insufficient to discriminate the build-up dynamics.

5.4 All-Metrics Overview

Figure 6 shows a radar chart summarising five representative metrics (AUC, AP, Macro-F1, mAP@0.3, mAP@0.7) across both models (3-seed mean); the full seven-metric breakdown is in Table 4. VideoMAE-B strictly dominates I3D on every axis, with the largest angular gap at AP and AUC and a visually striking improvement at the mAP vertices.

5.5 Discussion

Comparison caveat. Published SOTA rows in Table 3 use frame-level AUC evaluated on all 13 anomaly categories of UCF-Crime, while our rows use the 6-category violence subset with video-level AUC. These evaluations are *not* directly comparable; we include published numbers only to situate our work in the broader landscape. Within our controlled I3D-vs-VideoMAE-B comparison, the backbone swap yields +4.8 pp AUC and ×6.4 mAP@0.3—a strong signal regardless of evaluation differences.

Why localization improves most. VideoMAE's reconstruction pre-training forces the model to reason about spatiotemporal continuity across masked patches, producing features that encode temporal

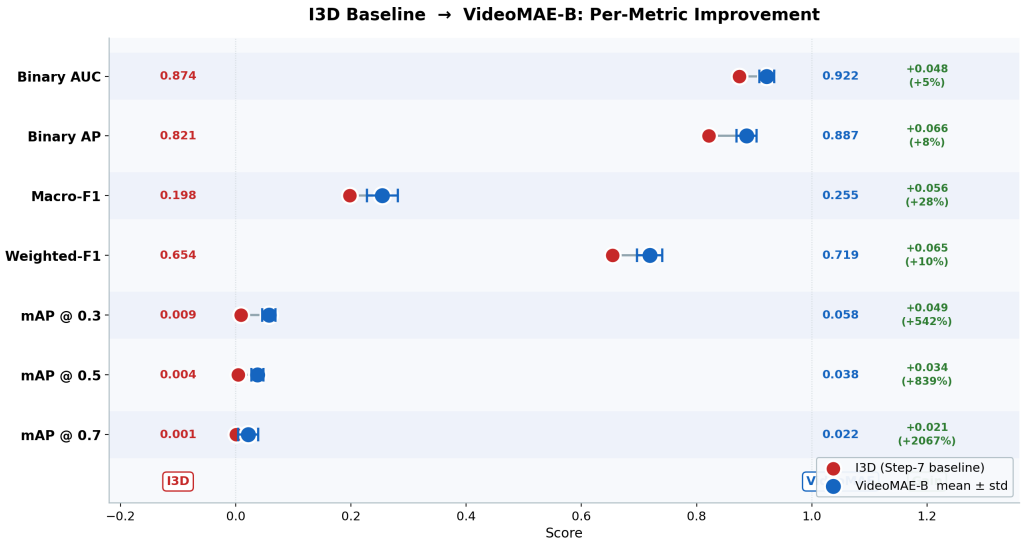

Figure 2: Per-metric improvement from I3D → VideoMAE-B (3-seed mean ± std). Green badges show absolute gain and relative percentage. Localization metrics (mAP) improve by an order of magnitude.

Table 5: Zero-shot cross-dataset transfer results. All external datasets are evaluation-only; model trained exclusively on UCF-Crime. † XD/UCF AUC ratio = 0.901 (passes ≥0.75 generalisation threshold).

Dataset	AUC	AP	Class metric	Outcome
UCF-Crime (seed-42 ref.)	0.927	0.868	Macro-F1 0.303	Reference checkpoint used for transfer
XD-Violence (zero-shot)†	0.836	0.885	Overlap Macro-F1 0.361	Binary transfer strong; class transfer partial (explosion F1 0.75, fighting F1 0.65)
RWF-2000 (fight val.)	0.862	0.819	Fight F1 0.246 (rec 0.14)	Ranking transfers; fight recall conservative
ShanghaiTech (robustness)	0.371	0.217	binary only (FN = 43/44)	Negative — non-violence domain shift

transitions more precisely than I3D's optical-flow stream. These richer temporal representations may contribute to sharper event boundary predictions, which would explain the disproportionate mAP gain at strict IoU thresholds—though a dedicated boundary-head ablation is needed to isolate this effect conclusively.

Cross-dataset transfer interpretation. XD-Violence binary transfer (ratio 0.901) confirms that VideoMAE-B violence representations generalise well across camera domains when the anomaly semantics overlap. RWF-2000's strong AUC (0.862) but weak fight recall (0.14) points to a calibration gap: the decision threshold is tightly tuned to UCF-Crime's surveillance footage and penalises RWF's shorter, close-up clips. ShanghaiTech's negative result (AUC 0.371) is expected by design—non-violence anomalies are out-of-distribution for a violence-trained model.

Overall performance context. The binary detection results are strong in absolute terms: AUC 92.2% and AP 88.7% are competitive with fully-supervised approaches on comparable subsets, and the localization gains ($\times$6.4–$\times$21.7 mAP) are substantial for a weakly-supervised system. The one area where absolute performance is limited is Macro-F1 (0.255), but this is a known artifact of severe class imbalance and scarce test samples for rare categories (e.g., only 2 Abuse test videos), not a failure of the VideoMAE-B representation itself.

Failure cases. The Fighting class collapse and the gap between Macro-F1 (0.255) and Weighted-F1 (0.719) reveal that rare categories remain challenging. Fighting shares motion statistics with normal crowded scenes, and 16-frame context appears insufficient to capture the build-up dynamics that distinguish a fight from ordinary movement; longer context windows or hierarchical aggregation are natural remedies. The RWF conservative recall further suggests calibration-aware thresholding as a priority for future work.

6 Conclusion

We demonstrated that replacing frozen I3D features with frozen VideoMAE-B features in a weakly-supervised RTFM + TRN + Boundary

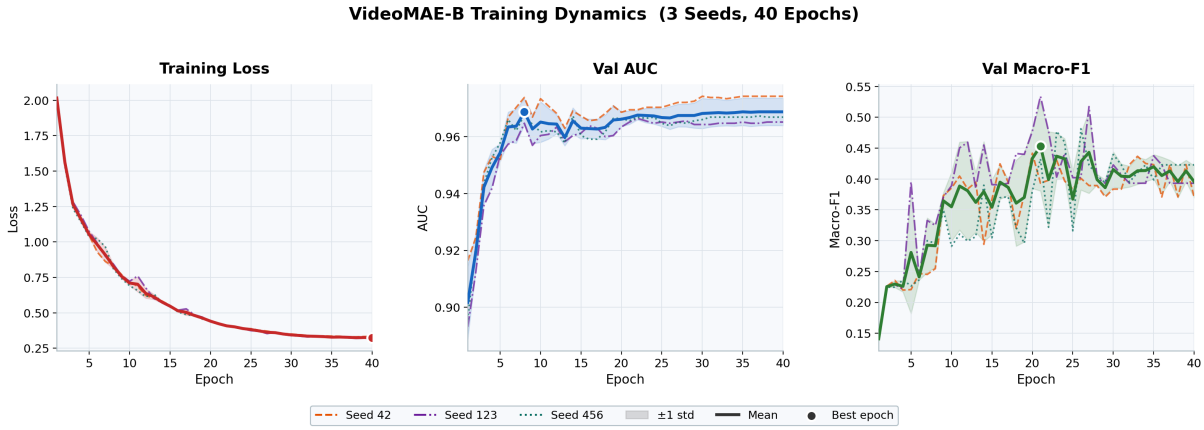

Figure 3: Training dynamics over 40 epochs (3 seeds, mean $\pm$ std band). **Left:** training loss—smooth convergence by epoch 15. **Center:** validation AUC—stable plateau above 91% from epoch 10. **Right:** validation Macro-F1—peaks near epoch 20.

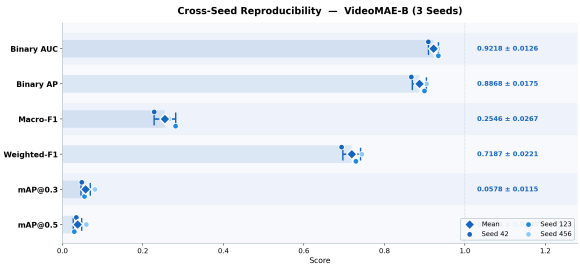

Figure 4: Cross-seed reproducibility. Diamond ($\diamond$) = mean $\pm$ 1 std; circles = individual seeds. Binary metrics are highly stable; localization metrics show higher but acceptable variance.

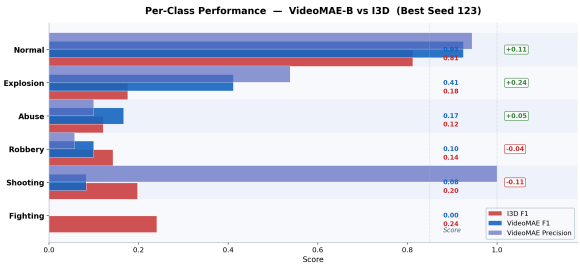

Figure 5: Per-class F1—I3D (red) vs. VideoMAE-B (blue), best seed 123. Delta badges show gain (green) or regression (red) vs. I3D.

Head pipeline yields consistent, large improvements in violence detection and temporal localization on the UCF-Crime violence subset (+4.8 pp AUC, $\times$6.4 mAP@0.3). Zero-shot transfer to XD-Violence remains strong for binary ranking (AUC 0.836, transfer ratio 0.901), with partial class-level transfer for explosion and fighting. RWF-2000 fight validation confirms useful anomaly ranking (AUC 0.862) but reveals conservative threshold behaviour under domain shift. ShanghaiTech

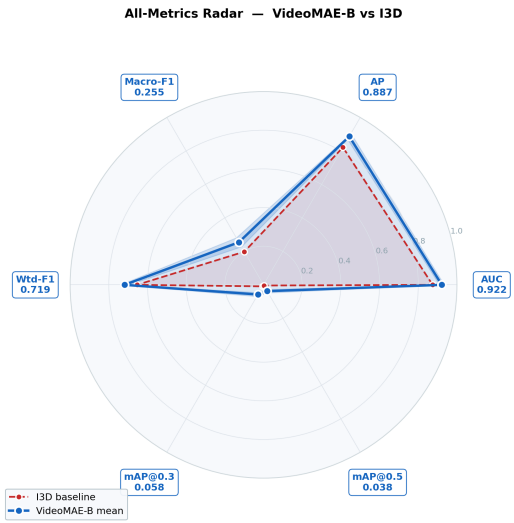

Figure 6: All-metrics radar—VideoMAE-B (blue, 3-seed mean) vs. I3D baseline (red dashed). VideoMAE-B dominates every axis.

robustness is negative (AUC 0.371), confirming that violence-trained representations do not generalise to non-violence anomaly semantics without retraining. A lightweight training recipe—inverse-frequency class weights, confidence-thresholded pseudo-labels, and cosine annealing LR—further stabilises UCF-Crime results across seeds.

Limitations and future work. Macro-F1 remains low (25.5%), driven primarily by the Fighting class collapse under 16-frame segmentation. Future directions include: (i) calibration-aware thresholding to improve external recall without increasing false positives, (ii) longer-context windows (32 or 64 frames) for fight-build-up capture, (iii) VideoMAE-Large backbone for richer

representations, and (iv) domain adaptation for non-violence anomaly benchmarks such as ShanghaiTech.

Team Contributions

Srinivasa Sai Chava conceived and led the entire project. He designed and implemented the full pipeline from scratch: UCF-Crime violence-subset data preparation and class-balanced splits; frozen I3D and VideoMAE-B feature extraction and caching; the RTFM-style weakly-supervised MIL scorer; the event-classification head with confidence-thresholded pseudo-label generation; the TRN temporal-refinement module; the boundary-confidence head and tIoU-based decode-time refinement; and the composite training objective with inverse-frequency class weighting and cosine-annealing LR scheduling. He ran all UCF-Crime training experiments (multi-seed sweeps, ablations), zero-shot cross-dataset evaluations on XD-Violence, RWF-2000, and ShanghaiTech, and the full interpretability package (error taxonomy, temporal-attention analysis, t-SNE feature-space visualisation, and boundary-precision analysis).

Yuxiang Liu contributed auxiliary analysis and evaluation tooling. He added threshold calibration and false-alarm analysis utilities, multi-run binary-accuracy and ablation reporting scripts, local experiment shell scripts, and extended evaluation entry points for multi-category UCF testing. He also extended the core boundary analysis, error taxonomy, evaluation, dataset, and training modules with additional logging and reporting hooks, added feature utility helpers, and authored the repository README and results-tracking document.

Vishakha –

Shristhy conducted exploratory data analysis on the ShanghaiTech and UCF-Crime datasets.

Anagha –

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